Score-Based Generative Modeling with Variance-Exploding SDEs: A Study of Reverse-Time Discretization

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Abstract

In this work, we implement a variance-exploding (VE) score-based generative model on the MNIST dataset and study the numerical behavior of reverse-time sampling. We train a neural network via denoising score matching to approximate the time-dependent score function, and generate samples using the Euler–Maruyama discretization of the reverse SDE. We analyze the effect of discretization step size and diffusion scale on sample quality, demonstrating that finer time discretization significantly improves reconstruction, and that mismatched noise scales degrade performance. These results illustrate the impact of numerical approximation in SDE-based generative modeling.

1 Introduction

Score-based generative models are a powerful framework for probabilistic modeling by transforming simple noise distributions into complex data distributions. In this project, we examine the numerical aspects of SDE-based sampling, particularly the role of discretization error in the Euler–Maruyama scheme.

2 Mathematical Background

2.1 Diffusion Processes and Forward SDE

Let $(W_t)_{t\in[0,T]}$ be a d-dimensional Brownian motion, and let $(x(t))_{t\in[0,T]} \in \mathbb{R}^d$ be the solution to the Itô stochastic differential equation (SDE)

$$dx = f(x,t)\,dt + g(t)\,dW_t, \tag{1}$$

where $f : \mathbb{R}^d \times [0,T] \to \mathbb{R}^d$ is the *drift* coefficient and $g : [0,T] \to \mathbb{R}$ is the *diffusion* coefficient. Under standard Lipschitz and growth conditions, we guarantee that this SDE admits a unique strong solution. Denote by $p_t(x)$ the probability density of $x(t)$, and by $p_{0t}(x(t) \mid x(0))$ the transition kernel of the process. The forward SDE satisfies that:

- $x(0) \sim p_0$, the data distribution,
- $x(T) \sim p_T$, the *prior* (noise) distribution (usually a tractable distribution such as the standard Gaussian distribution to allow easy sampling).

A common SDE is the *variance-exploding (VE)* SDE, obtained by setting $f = 0$, so

$$dx = g(t)\,dW_t, \quad \text{with} \quad g(t) = \sigma^t, \tag{2}$$

which perturbs data with increasing noise.

2.2 Reverse-Time SDE

A key result due to Anderson [2] states that the time-reversal of a diffusion process is a diffusion process; namely, the SDE:

$$dx = \left[f(x,t) - g(t)^2 \nabla_x \log p_t(x)\right] dt + g(t)\, d\bar{W}_t, \tag{3}$$

where $\bar{W}_t$ is a standard Wiener process evolving backward in time. Thus, if we have the *score function* $\nabla_x \log p_t(x)$ for all $t \in [0,T]$, we can simulate this reverse-time SDE starting from $x(T) \sim p_T$ to obtain samples from p_0. In the VE SDE case, this simplifies to

$$dx = -g(t)^2 \nabla_x \log p_t(x)\, dt + g(t)\, d\bar{W}_t. \tag{4}$$

2.3 Score-Based Modeling via Denoising Score Matching

To approximate the unknown score $\nabla_x \log p_t(x)$ for each $t \geq 0$, we train a neural network $s_\theta(x,t)$ by choosing $x(0) \sim p_0$ and $x(t) \sim p_{0t}(x(t) \,|\, x(0))$, and considering the objective

$$\theta^* = \arg\min_\theta \; \mathbb{E}_t \left[\lambda(t)\, \mathbb{E}_{x(0)} \mathbb{E}_{x(t)\,|\,x(0)} \left\| s_\theta(x(t),t) - \nabla_{x(t)} \log p_{0t}(x(t) \,|\, x(0)) \right\|^2\right], \tag{5}$$

where $\lambda(t) > 0$ is a weighting function. Under sufficient model capacity, the optimal solution satisfies

$$s_\theta^*(x,t) = \nabla_x \log p_t(x) \quad \text{for almost all } (x,t). \tag{6}$$

2.4 Gaussian Perturbation

For the variance-exploding SDE, the data is corrupted by Gaussian noise at time $t \geq 0$; that is:

$$x(t) = x(0) + \sigma(t) z, \quad z \sim \mathcal{N}(0, I), \tag{7}$$

where $\sigma(t)$ controls the noise level. The corresponding score has the closed form:

$$\nabla_{x(t)} \log p_{0t}(x(t) \,|\, x(0)) = -\frac{z}{\sigma(t)}. \tag{8}$$

Substituting this into the training objective, we obtain the loss used in implementation:

$$\mathbb{E}\left[\left\| s_\theta(x(t),t)\, \sigma(t) + z \right\|^2\right]. \tag{9}$$

This objective trains the model to predict the noise added to the data, which is equivalent to learning the score function.

3 Implementation Details

Our implementation is informed by the official codebase provided by Song et al. [3], adapted to a simplified MNIST setting. We implement a variance-exploding (VE) score-based generative model in PyTorch. The score $\nabla_x \log p_t(x)$ is approximated via denoising score matching, and sampling is performed by numerically solving the reverse-time SDE using the Euler–Maruyama method.

3.1 Dataset and Training

The model is trained on MNIST (28×28 grayscale images) normalized to $[0, 1]$, using mini-batches of size 32. Optimization is performed with Adam (learning rate 10^{-4}) for 15 epochs.
At each iteration, we sample $t \sim \text{Uniform}(\varepsilon, 1)$ with $\varepsilon = 10^{-5}$ and perturb data via

$$x(t) = x + \sigma(t)z, \quad z \sim \mathcal{N}(0, I). \tag{10}$$

The network is trained using the denoising score matching objective

$$\mathbb{E}\left[\|s_\theta(x(t), t)\,\sigma(t) + z\|^2\right]. \tag{11}$$

3.2 Network Architecture

The score network $s_\theta(x, t)$ is a U-Net style convolutional model with time conditioning. The time variable t is embedded using Gaussian Fourier features and injected via a small MLP. The architecture consists of residual blocks with group normalization and SiLU activation, arranged in an encoder–decoder structure with skip connections (two downsampling and two upsampling stages). The output is scaled as

$$\hat{s}_\theta(x, t) = \frac{\text{network}(x, t)}{\sigma(t)}, \tag{12}$$

to maintain stability across noise levels.

3.3 Sampling via Euler–Maruyama

Samples are generated by numerically simulating the reverse-time SDE using the Euler–Maruyama scheme. Starting from Gaussian noise $x_T \sim \mathcal{N}(0, I)$, the state is updated as

$$x_{k+1} = x_k + g(t_k)^2 s_\theta(x_k, t_k)\Delta t + g(t_k)\sqrt{\Delta t}\, z_k, \tag{13}$$

where $z_k \sim \mathcal{N}(0, I)$ and t_k decreases from T to 0.
This scheme approximates the continuous reverse diffusion process using discrete time steps. As a first-order method, it introduces discretization error that depends on the step size Δt: using fewer steps leads to a coarser approximation and can degrade sample quality. To study this effect, we vary the number of steps between 50 and 500.

3.4 Ablation Setup

We evaluate the sensitivity of sampling to:

- Number of discretization steps $\in \{50, 200, 500\}$
- Diffusion scale $\sigma \in \{10, 25, 50\}$

We train the model once and reuse it across all configurations, so variations in σ should be interpreted as robustness tests rather than matched evaluations.

4 Experiments

We evaluate the implemented variance-exploding score-based model on MNIST, focusing on the numerical behavior of reverse-time sampling. In particular, we study how sample quality depends on the discretization of the reverse SDE and the diffusion scale used during sampling.

4.1 Number of Discretization Steps

We vary the number of Euler–Maruyama steps used to approximate the reverse SDE. As a first-order scheme, Euler–Maruyama introduces discretization error for large step sizes, which can degrade sample quality. We therefore compare samples across different step counts while keeping all other components fixed.

4.2 Noise Scale σ

We also examine the effect of the diffusion scale σ, which controls the noise magnitude in the forward process and influences the reverse-time dynamics. Samples are generated for multiple values of σ to assess sensitivity to this parameter.

Remark. The model is trained once and reused across all sampling configurations. Thus, varying σ at sampling time constitutes a robustness analysis rather than a matched evaluation.

5 Results

Figure 1 shows the training loss over 15 epochs. The loss decreases rapidly during the initial epochs and then gradually stabilizes, which indicates that the score network successfully learns to approximate the time-dependent score function.

To analyze the effect of numerical discretization, Figures 2a, 2b, and 2c present samples generated with $\sigma = 25$ using 50, 200, and 500 reverse-time Euler–Maruyama steps.

With only 50 steps, the generated samples are dominated by noise, and the underlying digit structure is barely discernible. This indicates that such discretization leads to significant approximation error in the reverse SDE.

At 200 steps, we begin to recognize digit patterns, although the samples remain noisy. This suggests that the numerical approximation is improving but is still insufficient to fully capture the reverse-time dynamics.

At 500 steps, the generated digits are substantially clearer and more coherent. This demonstrates that increasing the number of discretization steps improves the fidelity of the reverse SDE approximation.

These observations are consistent with the theoretical expectation that the Euler–Maruyama method introduces discretization error that decreases as the step size Δt becomes smaller.

Finally, experiments with varying σ indicate that the sampling process is sensitive to the choice of diffusion scale. However, since the model is trained with a fixed σ and reused for different values during sampling, these results should be interpreted as a robustness analysis rather than a matched evaluation. In particular, smaller values of σ tend to produce unstable or noisy samples due to mismatch with the training distribution.

6 Discussion

The results highlight the trade-off between computational cost and numerical accuracy in reverse-time diffusion sampling. Increasing the number of Euler–Maruyama steps reduces discretization error and significantly improves sample quality, consistent with the first-order nature of the method.

The sensitivity to the diffusion scale σ further illustrates the importance of consistency between the training and sampling distributions. Since the model is trained with a fixed noise schedule, deviations in σ at sampling time lead to degraded performance, particularly for smaller values.

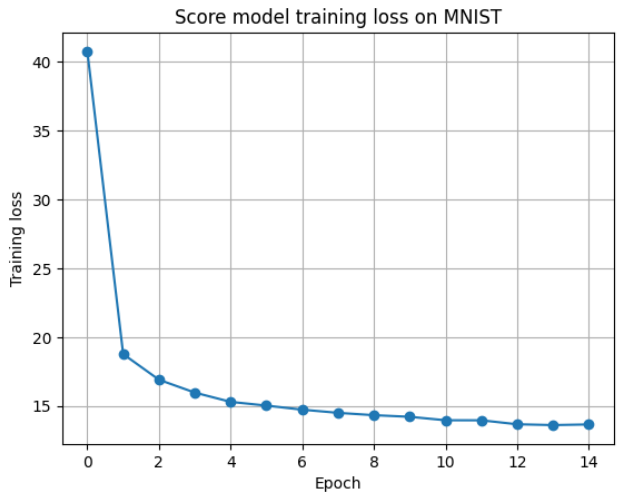
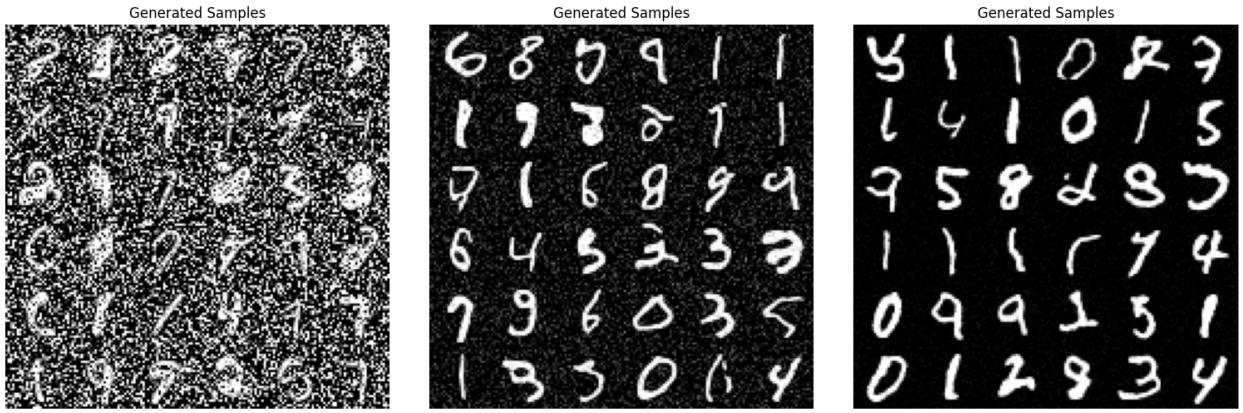


Figure 1: Training loss of the score network on MNIST over 15 epochs. The rapid initial decrease followed by gradual stabilization indicates successful optimization.



(a) 50 steps (b) 200 steps (c) 500 steps

Figure 2: Effect of the number of Euler–Maruyama steps on sample quality for $\sigma = 25$. Increasing the number of steps improves the approximation of the reverse SDE and leads to clearer samples.

While the Euler–Maruyama scheme provides a simple and effective baseline, more advanced samplers (e.g., predictor–corrector methods) could further improve sample quality without requiring substantially more computation.

7 Conclusion

We implemented a score-based generative model using a variance-exploding SDE and investigated the numerical properties of reverse-time sampling. The results demonstrate that sample quality strongly depends on the discretization of the reverse SDE, with finer step sizes yielding clearer and more coherent samples. Our experiments also show that the choice of diffusion scale affects stability, particularly under mismatch between training and sampling configurations.

References

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[3] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-Based Generative Modeling through Stochastic Differential Equations (Official Implementation)," GitHub repository, 2021. `https://github.com/yang-song/score_sde`